

SDR/ML Algorithm Sandbox

explore-scaffold-notebook-version

September 14, 2026

What Is the Sandbox?

The **sandbox** is an integrated pipeline where a team can:

1. Express an algorithm idea (pseudocode or free-form).
2. Generate a runnable Jupyter notebook automatically.
3. Follow a documented structure so every notebook is consistent.
4. Audit the notebook for correctness before merging.
5. Version-control everything through a disciplined git flow.

It combines **five tools** into a single workflow:

Tool	Role
pseudocode-general	High-level flow description (language-agnostic)
pseudocode-specific	Detailed algorithm design with control flow
python-notebook	Generates valid notebook JSON from a description
project-scaffold	3-layer documentation for agent navigation
git-flow	Branch strategy, PR workflow, conventional commits

Git Flow — Conceptual View

Branch Hierarchy

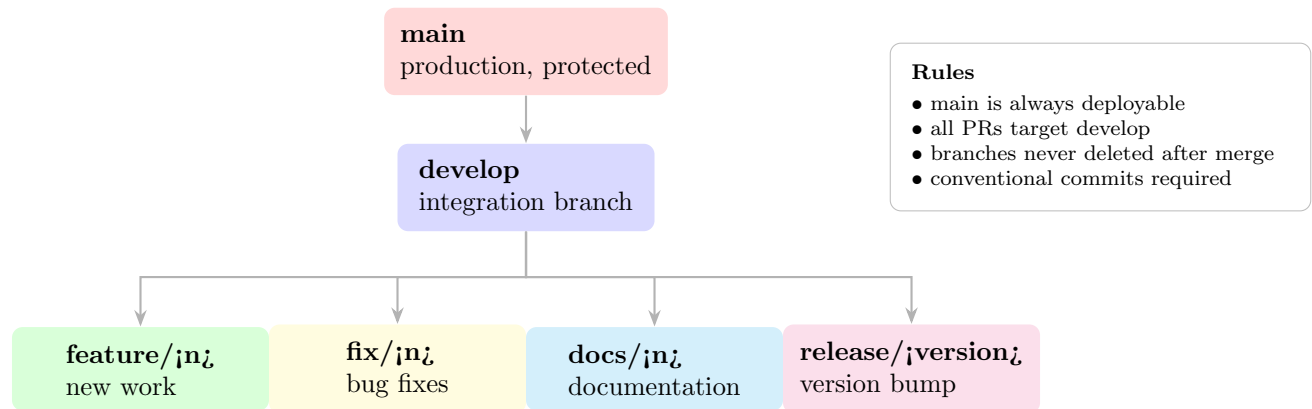


Figure 1: Git Flow — branch hierarchy.

Branch Lifecycle

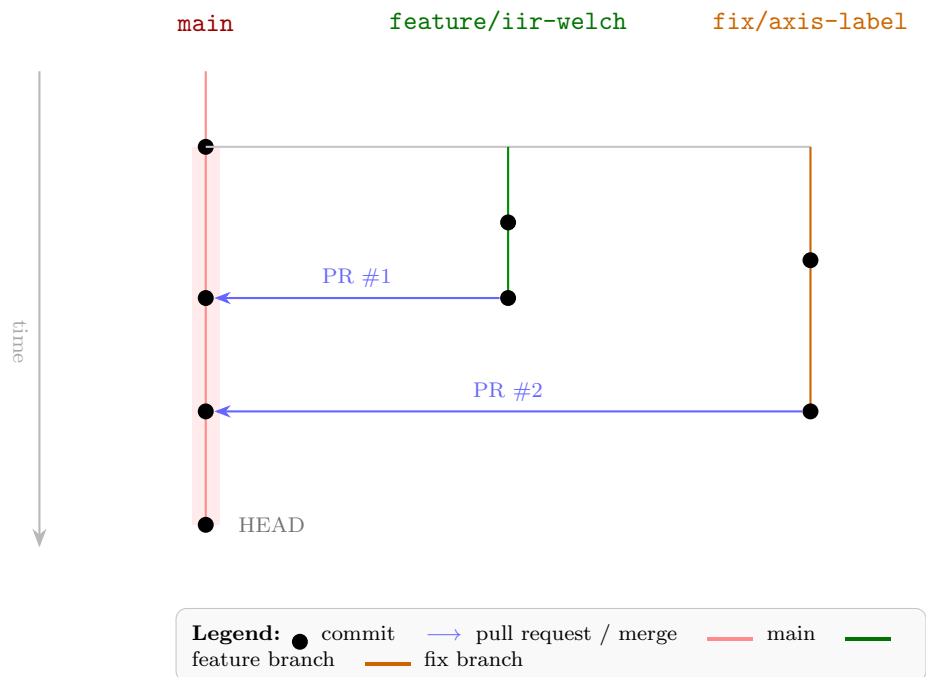


Figure 2: Git Flow — parallel branches with PR merges into main.

Flow Sandbox 1 — Pseudocode-Driven

Entry: User provides general pseudocode. **Exit:** Notebook merged via git flow.

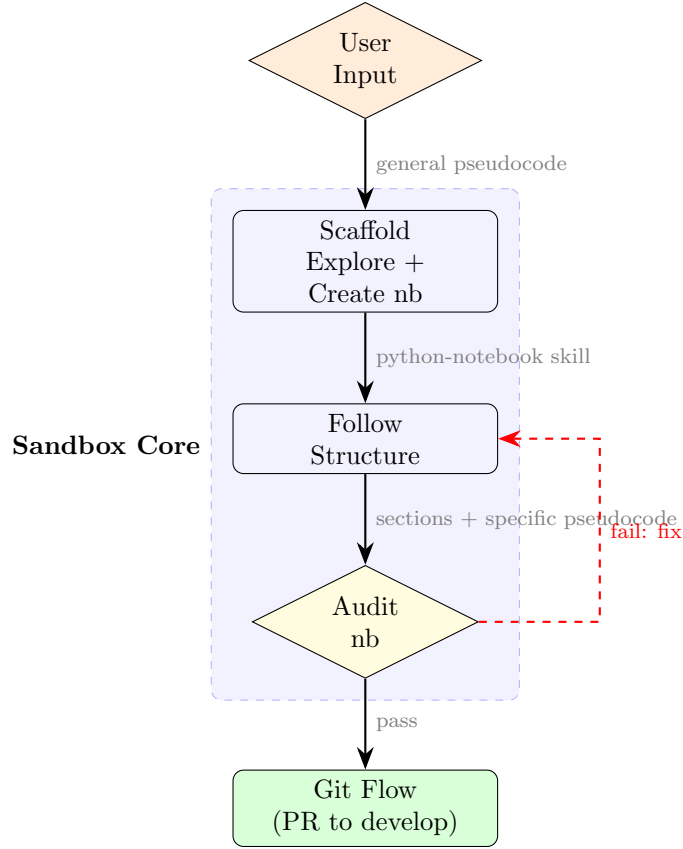


Figure 3: Flow Sandbox 1 — pseudocode-driven pipeline.

Steps:

1. **User Input** — Provide general pseudocode (e.g., SDR monitoring pipeline).
2. **Scaffold Explore + Create nb** — Agent explores scaffold documentation, generates notebook using python-notebook skill.
3. **Follow Structure** — Notebook sections follow the scaffold; each section has its own specific pseudocode.
4. **Audit nb** — Validate: notebook format, axis conventions (N, 2, L), power calculations, missing cells.
5. **Git Flow** — Create feature branch, commit with conventional message, open PR to develop.

Conceptual Example

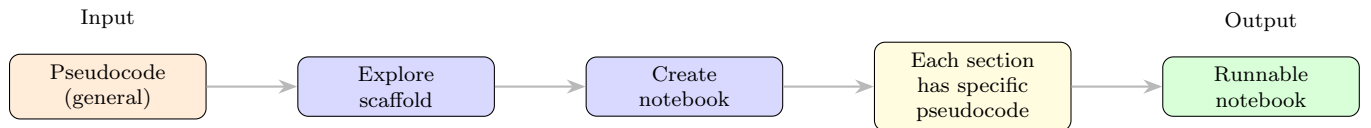


Figure 4: Flow Sandbox 1 — from pseudocode to runnable notebook.

Flow Sandbox 2 — Idea-Driven

Entry: Team member has a test idea (no pseudocode yet). **Exit:** Notebook merged via git flow.

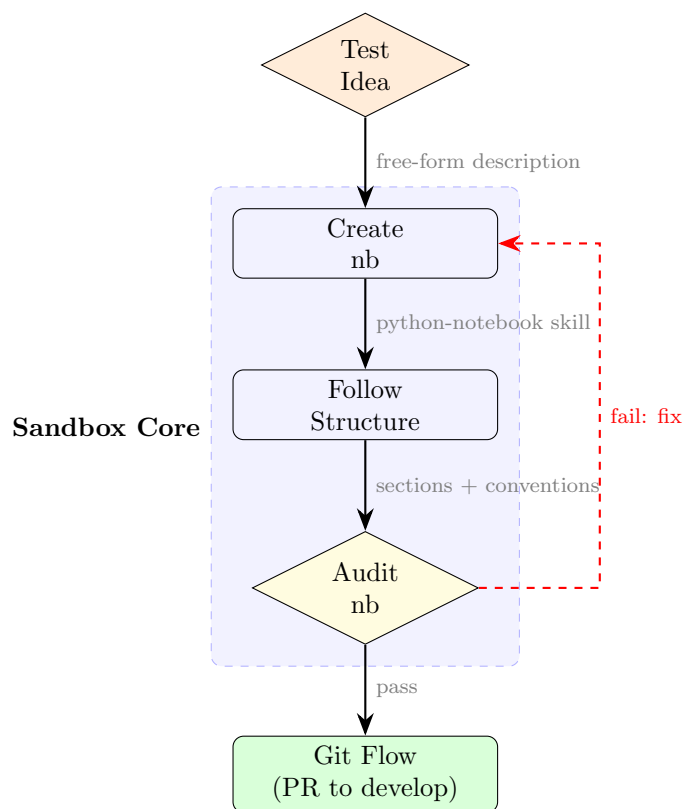


Figure 5: Flow Sandbox 2 — idea-driven pipeline.

Steps:

1. **Test Idea** — Team member describes what they want to test (e.g., compare Welch vs periodogram for weak signals).
2. **Create nb** — Agent generates notebook using python-notebook skill from the description.
3. **Follow Structure** — Notebook follows scaffold conventions: IQ shape (N, 2, L), axis explanations, power formulas.
4. **Audit nb** — Validate format, conventions, missing cells, correctness.
5. **Git Flow** — Branch, commit, PR to develop.

Conceptual Example

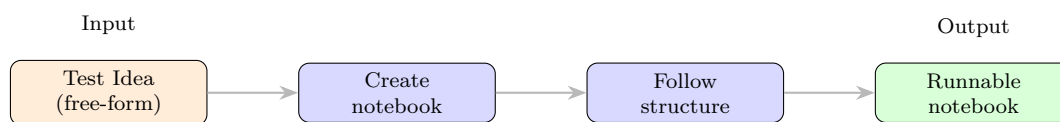


Figure 6: Flow Sandbox 2 — from idea to runnable notebook.

What Is Missing

The sandbox is functional but incomplete. These components need to be built:

1. Dockerization

Reproducible environment for all team members.

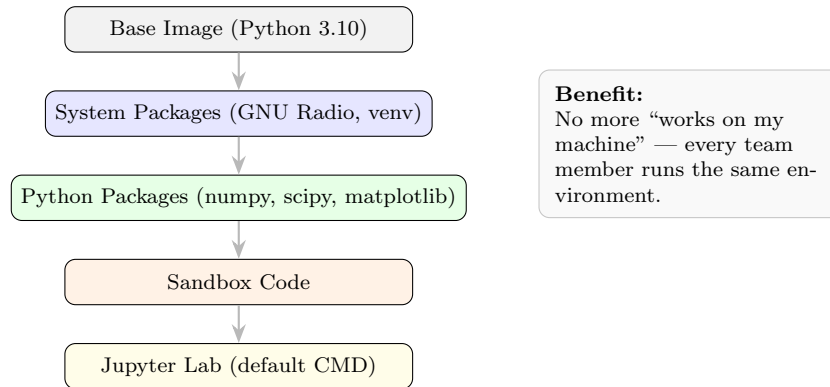


Figure 7: Dockerization — layered container structure.

2. Dataset Generation (Simulated)

Synthetic IQ data for testing without hardware.

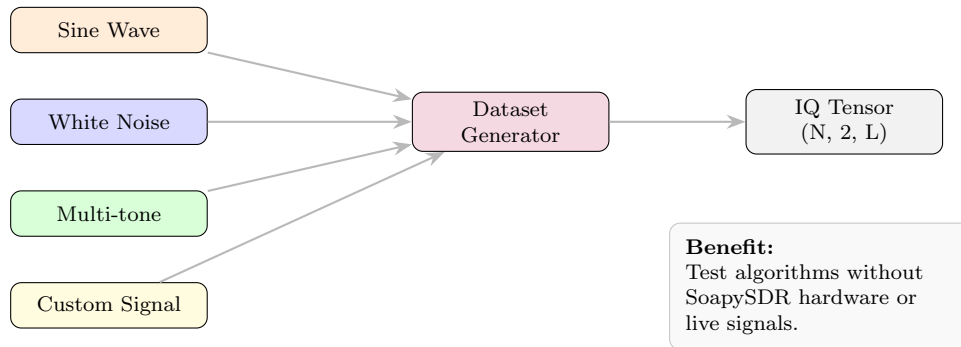


Figure 8: Dataset Generation — synthetic IQ signal types.

3. SoapySDR Skill

Skill for interacting with SDR hardware via SoapySDR.

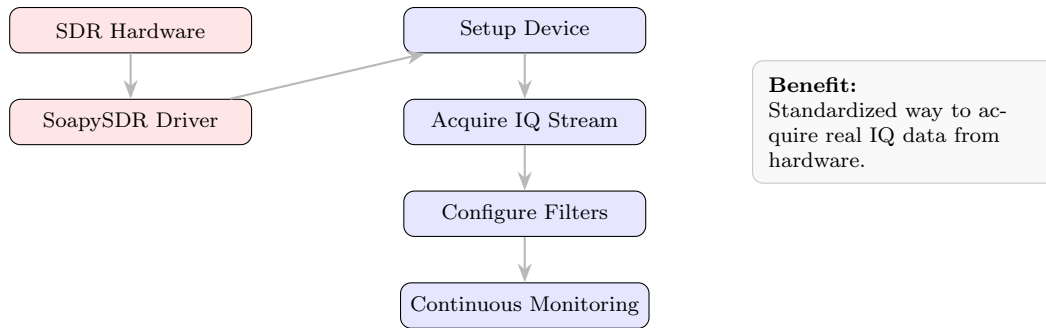


Figure 9: SoapySDR Skill — hardware to software interface.

4. Audit Skill

Automated notebook validation.

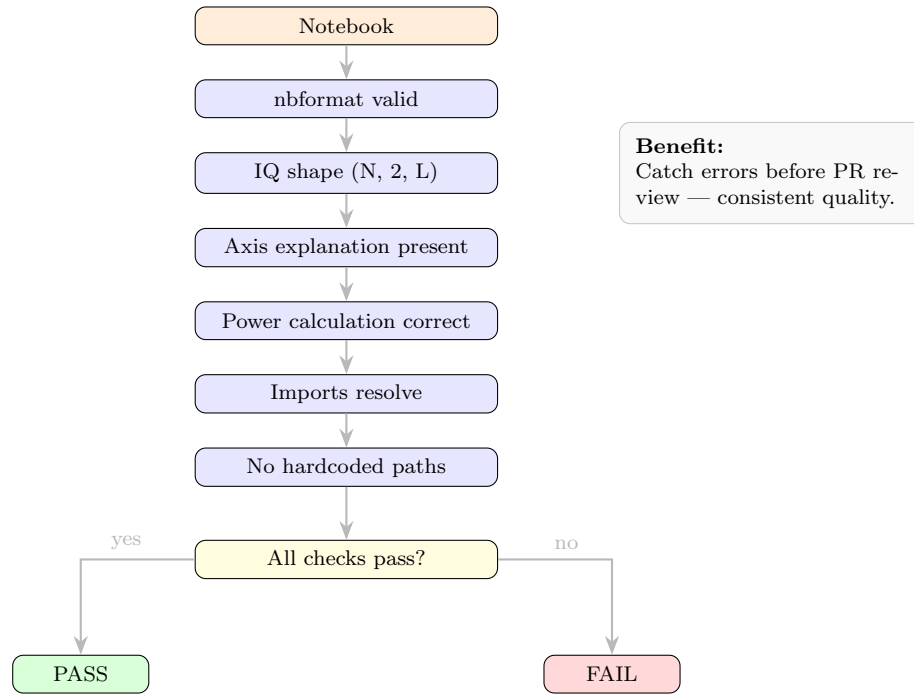


Figure 10: Audit Skill — automated validation checklist.

5. Harness Logic / MCP Server

Orchestrate the full pipeline as a single command.

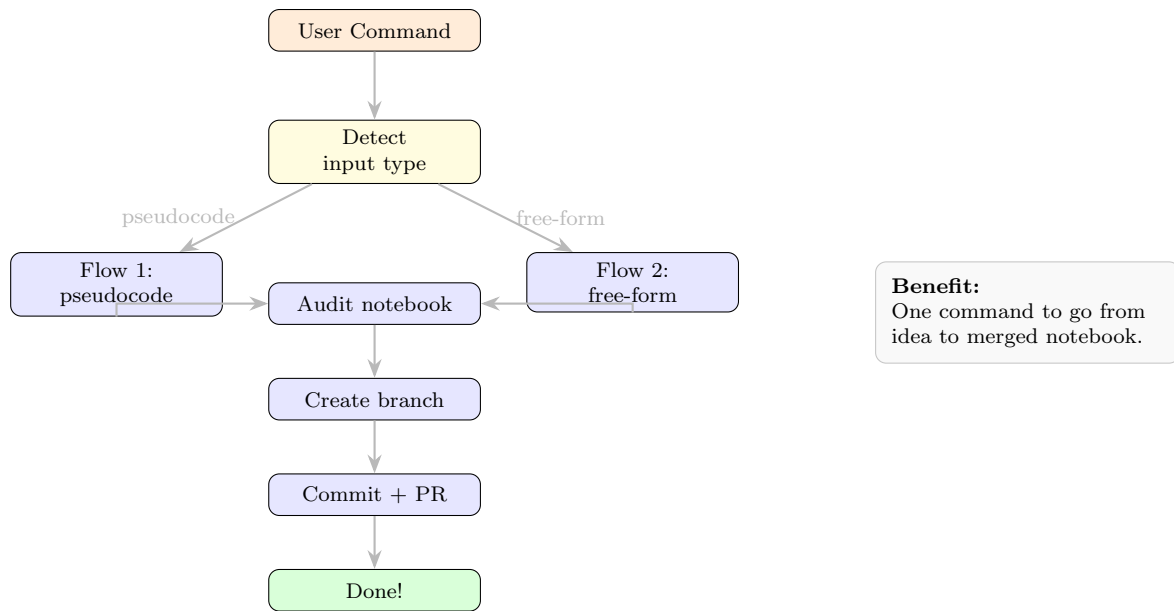


Figure 11: Harness — single-command orchestration.

Sandbox Architecture — Full View

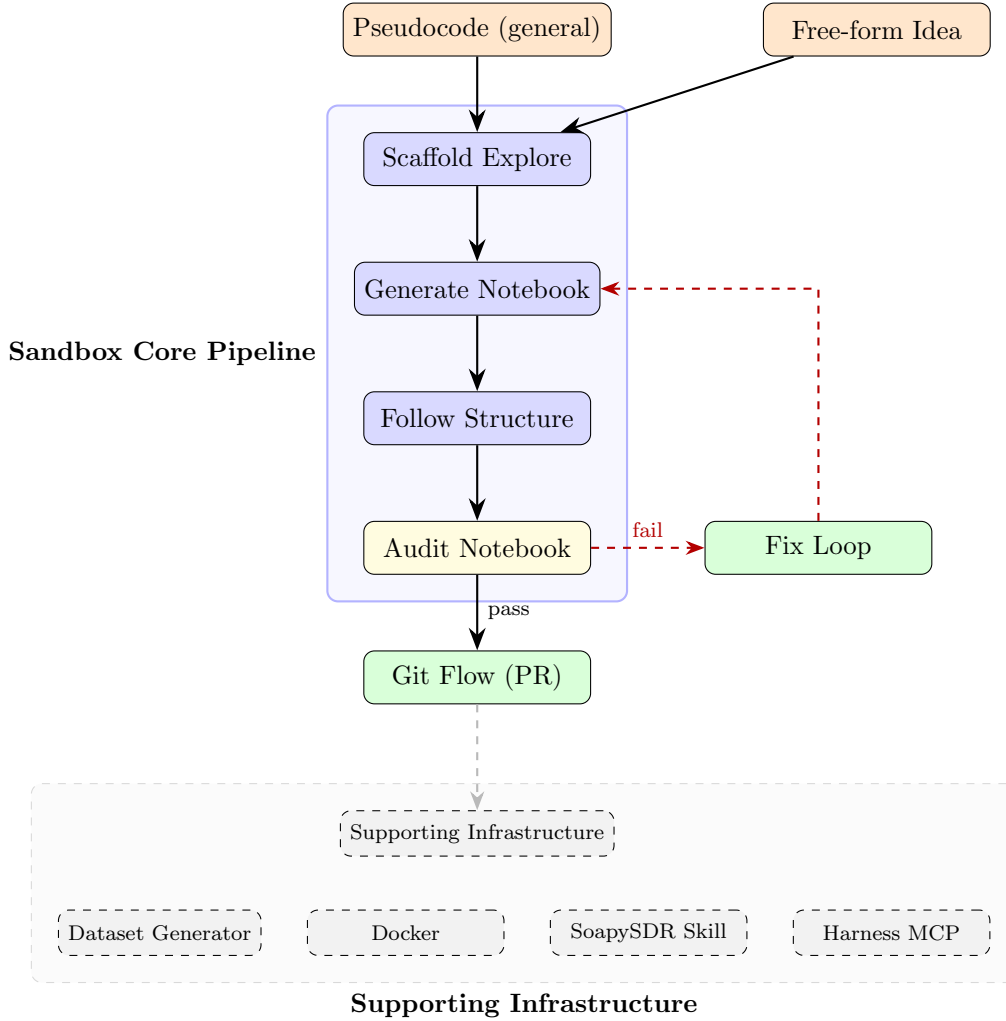


Figure 12: Sandbox architecture — core pipeline and supporting infrastructure.

Summary

Component	Status	Purpose
pseudocode-general	Done	High-level flow description
pseudocode-specific	Done	Detailed algorithm design
python-notebook	Done	Generate notebook from description
project-scaffold	Done	3-layer documentation navigation
git-flow	Done	Branch strategy, PR workflow
Dockerization	Missing	Reproducible team environment
Dataset Generation	Missing	Synthetic IQ data for testing
SoapySDR Skill	Missing	Hardware IQ acquisition
Audit Skill	Missing	Automated notebook validation

Component	Status	Purpose
Harness / MCP	Missing	Orchestrate full pipeline